ASR Shootout for Indian Conversational Speech

Benchmarking Deepgram, Whisper Large-v3, and Sarvam AI for Locality Recognition

Candidate: Sagnick Das

Role: AI Intern Assessment

---

1. Introduction

This project evaluates Automatic Speech Recognition (ASR) systems for Indian conversational speech in real-world conditions. The primary goal was not only to measure transcription quality, but also to evaluate whether these systems can reliably capture locality names from noisy, informal, multilingual speech.

This matters directly for voice-based hiring platforms where users interact over phone calls or WhatsApp voice notes. In such systems, ASR failure is rarely about full-sentence understanding. The actual business failure happens when critical entities such as locality names, job roles, or salary expectations are transcribed incorrectly.

The evaluation was performed using a custom dataset of self-recorded Bangalore locality utterances under varied acoustic conditions. All models were benchmarked against Deepgram, which was used as the baseline as required in the assignment.

The three systems evaluated were:

Model	Type	Reason for Selection
Deepgram Nova-2	API-based commercial ASR	Baseline model with strong production reputation
Whisper Large-v3	Open-source multilingual ASR	State-of-the-art open-source benchmark model
Sarvam AI Saarika	Indic-focused ASR API	Optimized specifically for Indian languages and accents

The benchmark focused on both transcription quality and production practicality.

---

2. Dataset and Evaluation Setup

2.1 Custom Conversational Dataset

A custom dataset was recorded consisting of 20 conversational utterances containing Bangalore locality names such as Koramangala, Whitefield, HSR Layout, Marathahalli, and Rajajinagar.

Instead of reading isolated words, the recordings were embedded naturally inside conversational sentences such as:

> “Haan, main Koramangala mein rehta hoon.”

This was done intentionally because production traffic is conversational, not studio-clean keyword audio.

The recordings were intentionally varied across multiple acoustic conditions:

Condition	Description
Clean	Quiet indoor recording
Street Noise	Traffic/background outdoor noise
Phone Call	Telephony-compressed audio
Whispered	Low-energy speech
Rushed	Fast conversational delivery

The recordings were captured using a phone microphone rather than professional audio equipment to better simulate real deployment conditions.

2.2 Metrics Used

Traditional Word Error Rate (WER) alone is insufficient for this task.

For example:

> “Main Koramangala mein rehta hoon”

and

> “Main Kormangla mein rehta hoon”

may produce a relatively low WER penalty while still completely breaking downstream entity extraction.

Because of this, the evaluation used multiple metrics:

Metric	Purpose
WER	Measures word-level transcription accuracy
CER	Measures character-level transcription accuracy
LEA Exact	Exact locality extraction accuracy
LEA Fuzzy	Fuzzy locality matching accuracy
Latency	Measures inference responsiveness
Cost	Important for production scalability

LEA (Locality Extraction Accuracy) was introduced as the most business-relevant metric because the assignment centers around correct recognition of locality names.

---

3. Benchmark Results

3.1 Overall Performance

Model	WER ↓	CER ↓	LEA Exact ↑	LEA Fuzzy ↑	Avg Latency
Deepgram Nova-2	Moderate	Moderate	Good	Very Good	Very Fast
Whisper Large-v3	Best	Best	Best	Best	Slow
Sarvam AI Saarika	Good	Good	Strong	Strong	Fast

3.2 Key Observations

Whisper Large-v3

Whisper Large-v3 achieved the strongest transcription quality overall, especially under noisy and code-switched conditions. It handled Hinglish and conversational phrasing significantly better than expected.

However, the model is extremely resource-heavy. Running large-v3 on constrained hardware becomes difficult because the TensorFlow/PyTorch runtime itself consumes substantial RAM before inference even begins.

This creates deployment friction for low-memory environments.

Deepgram Nova-2

Deepgram performed very well in terms of latency and responsiveness. The API was simple to integrate and inference times were consistently fast.

However, Deepgram occasionally struggled with Indian locality names that are uncommon in western-centric datasets. Errors increased under noisy or rushed speech.

Its strength is production readiness and speed, not necessarily localization.

Sarvam AI Saarika

Sarvam AI performed surprisingly well for Indian speech despite being lighter than Whisper.

The model handled Indian accents and locality pronunciations more naturally than Deepgram in several cases. This suggests that region-specific optimization matters more than raw model scale in certain scenarios.

The tradeoff was occasional instability under highly noisy audio.

---

4. Failure Analysis

The most important part of this benchmark was understanding *where* models fail.

4.1 Locality Name Distortion

Rare locality names caused major issues across all systems.

Examples included:

Actual	Incorrect Predictions
Byatarayanapura	“Bayatarayanpur”, “Byatranpur”
Kadugondanahalli	“Kadugonda halli”, “Kadugondnahali”
Rajarajeshwarinagar	Fragmented into multiple tokens

These failures matter because downstream entity systems depend on correct locality recognition.

4.2 Noise Sensitivity

Street-noise recordings caused large degradation for all models.

Whisper remained the most robust overall, while Deepgram and Sarvam occasionally dropped entire words during heavy background traffic noise.

Phone-call compressed audio also introduced artifacts that reduced recognition accuracy.

4.3 Whispered Speech

Whispered recordings exposed a common weakness:

Low-energy speech caused models to hallucinate missing words or partially infer phrases from context.

This becomes dangerous in production because the transcription may appear fluent while still being factually incorrect.

4.4 Code-Switching

Hinglish utterances produced mixed outcomes.

Whisper handled Hindi-English transitions best due to multilingual training scale.

Deepgram occasionally transliterated Hindi words incorrectly into English phonetics.

Sarvam handled Indian pronunciation well but sometimes over-normalized English locality names.

---

5. Production Considerations

ASR selection is not only an accuracy problem.

In production systems, deployment constraints matter equally.

Factor	Deepgram	Whisper Large-v3	Sarvam AI
Setup Complexity	Low	High	Low
GPU Requirement	No	Yes (recommended)	No
Latency	Excellent	Slow	Good
Offline Support	No	Yes	Limited
Indic Speech Handling	Moderate	Excellent	Strong
Scaling Cost	Usage-based	Infrastructure-based	Usage-based

Whisper provides the best raw transcription quality but introduces significant operational cost.

For example:

- Large VRAM requirement
- High inference latency
- Heavy runtime dependencies

- Difficult deployment on low-memory environments

Deepgram is easier to scale operationally but sacrifices some localization quality.

Sarvam represents a strong middle ground for Indian deployments.

---

6. Recommendation

If accuracy is the top priority and GPU infrastructure is available, Whisper Large-v3 is the strongest overall ASR system in this benchmark.

However, for real production deployment in an Indian voice hiring platform, the best practical choice is likely a hybrid strategy:

- Use Deepgram or Sarvam for fast real-time inference
- Route difficult or low-confidence samples to Whisper for secondary processing
- Add a post-ASR entity correction layer using fuzzy matching for locality names

This hybrid approach balances:

- Accuracy
- Latency
- Infrastructure cost
- Scalability

A pure WER-focused evaluation would miss this conclusion entirely.

The benchmark also showed that entity extraction accuracy matters more than perfect sentence transcription in telephony hiring workflows.

---

7. Limitations

This evaluation still has several limitations:

- Small custom dataset size
- Limited speaker diversity
- No streaming ASR benchmark
- Limited multilingual coverage beyond Hinglish
- No rigorous throughput testing under concurrent load

Future work could include:

- Larger multi-speaker datasets
- Streaming latency benchmarks

- Domain-specific fine-tuning
- Post-ASR entity correction models
- Evaluation across additional Indian languages

---

8. Conclusion

This project demonstrated that evaluating ASR systems for Indian conversational speech requires more than comparing WER scores.

The benchmark revealed that:

- Whisper Large-v3 delivers the best raw transcription quality
- Deepgram offers the best operational simplicity and latency
- Sarvam AI provides strong localization advantages for Indian speech
- Locality extraction accuracy is more important than generic transcription quality
- Real-world noisy audio dramatically changes model behavior

The most important insight from this benchmark was that deployment constraints and entity-level accuracy matter just as much as model benchmark scores.

An ASR system that performs well on clean benchmark datasets may still fail in production if it cannot reliably recognize conversational Indian locality names under noisy telephony conditions.

That gap between benchmark performance and production behavior is exactly what this evaluation attempted to expose.